Algebra 1
Unit 2
Lesson 1 Practice

Name___Answer Key________
Date______________________
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1. The power of a power law of exponents states to keep the [○ base / ○ coefficient / ○ exponent] and then [○ add / ○ subtract / ○ multiply] the powers.

Selected: base; multiply

2. Diana simplified the algebraic expression and made an error. Find and fix the mistake.

$$(-3x^3y^5)^3(1.5xy^0)$$
$$(-3x^9y^{15})(1.5x)$$
$$(-3y \cdot 1.5)(y^{15})(x^9 \cdot x)$$
$$-4.5x^9y^{15}$$

$$(-3x^3y^5)^3(1.5xy^0)$$
$$(-27x^9y^{15})(1.5x)$$
$$-40.5x^{10}y^{15}$$

3. Write a note to Diana explaining the error she made in the previous question.
Answers will vary. Remember, Diana, the power of a power law says to multiply all powers. This includes the leading coefficient. In the term $(-3x^3y^5)^3, -3$ has a power of 1. So, the power of 3 must be multiplied to the power of 1 also. In the term x, there is always a power of 1. When there is none, there is one, x^1.

Determine an equivalent algebraic monomial expression for each expression using the laws of exponents. Show your work.

4. $(-8a^5b)(3ab^4)$

$(-8 \cdot 3)(a^5 \cdot a)(b \cdot b^4)$

$-24a^6b^5$

5. $(\frac{-1}{4}x^3y^2)(4x^8y^3)(-5y^2)$

$(\frac{-1}{4} \cdot 4 \cdot -5)(x^3 \cdot x^3)(y^2 \cdot y^3 \cdot y^2)$

$5x^{11}y^7$

6. $(2.8kj^9)(3.5k^8)(\frac{1}{3}k^5j^2)$

$(2.8 \cdot 3.5 \cdot \frac{1}{3})(k \cdot k^8 \cdot k^5)(j^9 \cdot j^2)$

$\frac{49}{15}k^{14}j^{11}$ or $3.2\bar{6}k^{14}j^{11}$

7. $(-3mn^5p^4)^3$

$-3^3m^3n^{15}p^{12}$

$-27m^3n^{15}p^{12}$

8. $(2.3c^4d)(-2.4c^{-3}d^0)(-c^2d^{-5})$

$(2.3 \cdot -2.4 \cdot -1)(c^4 \cdot c^{-3} \cdot c^2)(d \cdot d^0 \cdot d^{-5})$

$\frac{5.52c^3}{d^4}$

9. $(\frac{5}{6}r^{-8})^{-2}(-9.6r^5s^4)^0$

$(\frac{5}{6})^{-2}(r^{-8})^{-2} \cdot 1$

$(\frac{6}{5})^2 \cdot r^{16}$

$\frac{36}{25}r^{16}$ *or* $1.44r^{16}$

10. $(-8y^5z)^3(5y^7z^6)^3$

$-8^3y^{15}z^3 \cdot 5^3y^{21}z^{18}$
$-512y^{15}z^3 \cdot 125y^{21}z^{18}$
$-512 \cdot 125y^{36}z^{21}$

$-64{,}000y^{36}z^{21}$

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Lesson 2 Practice

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Find the quotient of each expression by using the laws of exponents to generate an equivalent expression.

1. $\frac{3x^6y^5}{12xy^8}$

$\frac{1x^5}{4y^3} = \frac{x^5}{4y^3}$

2. $\frac{2a^2b^7}{12a^3b^4}$

$\frac{1b^3}{6a} = \frac{b^3}{6a}$

3. $(\frac{-2s^3t^6}{8t^6})^5$

$\frac{-32s^{15}}{32768} = \frac{-s^{15}}{1024}$

4. $\frac{-2k^5j^9}{-6k^4j^{10}}$

$\frac{1k}{3j} = \frac{k}{3j}$

5. $\frac{28w^8z^6}{42w^{10}z^4}$

$\frac{2z^2}{3w^2}$

6. $\left(\frac{xy^9}{-5y^4}\right)^0$

1

7. $\frac{6r^8s^5}{-12r^5s^8}$

$$\frac{-1r^3}{2s^3} = \frac{-r^3}{2s^3}$$

8. $\frac{5x^5}{60x^{11}s^3}$

$$\frac{1}{12x^6s^3}$$

9. $\left(\frac{13x^0z^5}{-x^4z}\right)^2$

$$\frac{169 \cdot 1 \cdot z^{10}}{x^8z^2} = \frac{169z^8}{x^8}$$

10. $\left(\frac{-25r^{10}p^8}{-10r^4z}\right)^3$

$$\frac{125r^{18}p^{24}}{8z^3}$$

Algebra 1
Unit 2
Lesson 3 Practice

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1. Pam was simplifying the expression and made an error. Her steps are shown. Find and fix the mistake.

$$(\frac{4x^3y^3 \cdot 2xy}{3xy^2})^2$$

Step 1: $(\frac{8x^4y^4}{3xy^2})^2$

Step 2: $\frac{64x^6y^6}{9x^2y^4}$

Step 3: $\frac{64x^4y^2}{9}$

Step 1: $(\frac{8x^4y^4}{3xy^2})^2$

Step 2: $\frac{64x^8y^8}{9x^2y^4}$

Step 3: $\frac{64x^6y^4}{9}$

2. Write a note to Pam explaining her error.
Answers will vary.
When using the Power of a Power Exponent Law, multiply the exponent to all the powers inside the parentheses in both the numerator and denominator.

For questions 3 through 5, find an equivalent expression using the laws of exponents.

3. $\frac{14x^6y^6}{12x^{10}y^3}$

$\frac{7y^3}{6x^4}$

4. $\frac{4x^6y^6}{6x^2} \cdot \frac{3x}{4y}$

$\frac{12x^7y^6}{24x^2y} = \frac{x^5y^5}{2}$

5. $(\frac{4x^2y^3}{5x^0y^5})^2$

$\frac{16x^4y^6}{25y^{10}} = \frac{16x^4}{25y^4}$

6. Prove $2^{2x+4} = 16 \cdot 2^{2x}$ by using the laws of exponents.

$2^{2x} \cdot 2^4 = 2^{2x} \cdot 16 = 16 \cdot 2^{2x}$

7. Prove $4^{x-2} = \frac{4^x}{16}$ by using the laws of exponents.

$4^x \cdot 4^{-2} = \frac{4^x}{4^2} = \frac{4^x}{16}$

8. Write an equivalent expression for $(-3.1)^{4x} \cdot (-3.1)^{2x+7}$

$(-3.1)^{6x+7}$

9. Use the law of exponents to write an equivalent expression for 5^{7-3x}.

$\frac{78125}{5^{3x}}$

10. Use the law of exponents to write an equivalent expression for $(2.8)^{6x+8}$.

$3778.019983 \cdot (2.8)^{6x}$

Algebra 1
Unit 2
Lesson 4 Practice

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1. True or False? *The polynomial* $5xy^7 + 3x^4y^2 - x^2y^2$ *is written in standard form.*

True. The highest degree is written first then the terms are in order from highest degree to lowest degree.

2. Write the polynomial $-10xy^5 + 2x^3y^2 - 6x^2y^2$ in standard form.

$$-10xy^5 + 2x^3y^2 - 6x^2y^2$$

For questions 3-4, rewrite each polynomial in standard form, then identify the features of the polynomial.

3. $4y^2 - 8y^3 + 2 - 6y$

Standard Form: $-8y^3 + 4y^2 - 6y + 2$

Number of Terms: 4

Degree of the polynomial: 3

Leading Term: $-8y^3$

Leading Coefficient: -8

4. $-3mp^2 + 7m^2p^2 - 4p - 6m^4p^2$

Standard Form: $-6m^4p^2 + 7m^2p^2 - 3mp^2 - 4p$

Number of Terms: 4

Degree of the polynomial: 6

Leading Term: $-6m^4p^2$

Leading Coefficient: -6

5. Create a polynomial that has 3 terms, a degree of 6, a leading coefficient of 5, and a constant term of -1.8.

$5x^6 + 4x^2 - 1.8$ Answers will vary.

6. Alexis found the product of $3x^2(5x^4 - 3x^3 + 7x - 8)$ to be $15x^8 - 9x^6 + 21x^3 - 8x^2$. Is she correct? Explain.

No, she multiplied rather than adding exponents and multiplied the last term incorrectly.

7. Find the product of $-2r^2(12r^3 - 2r^2 - 9r + 11)$ in standard form.

$$-24r^5 + 4r^4 + 18r^3 - 22r^2$$

Find the product of each by distributing the monomial to the polynomial. Express the product in standard form.

8. $\frac{1}{2}y^2(6y^2 - 6y + 7)$

$$3y^4 - 3y^3 + 3.5y^2$$

9. $(6xy^2 + 4x^2y + 7xy + 8)(\frac{2}{3}x^3y^2)$

$$4x^4y^4 + \frac{8}{3}x^5y^3 + \frac{14}{3}x^4y^3 + \frac{16}{3}x^3y^2$$

10. $-2.8a^3(1.2a^4 - 2.3a^3 + 0.6a)$

$$-3.36a^7 + 6.44a^6 - 1.68a^4$$

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Lesson 5 Practice Period__________________

1. Are the terms $6ab^3$ and $-2a^3b$ like terms? Why or why not? Explain.

   No, they have unlike bases. They have same variables but different exponents.

2. Find the sum of $(8-3t-5t^2)$ and $(8t+3t^2)$.

   $(-5t^2-3t+8)+(3t^2+8t)=-2t^2+5t+8$

3. Write the sum from question 2 in standard form.

   $-2t^2+5t+8$

4. Find the difference of $(7.8n^2+2.5-2n)$ and $(-2n^2-8)$.

   $(7.8n^2+2.5-2n)-(-2n^2-8)=9.8n^2-2n+10.5$

5. Write the difference from question 4 in standard form.

   $9.8n^2-2n+10.5$

6. What polynomial must be added to $-3c^2+3-6c$ in order to get a sum of $2c^2-c+5$?

   $(2c^2-c+5)-(-3c^2+3-6c)=5c^2+5c+2$

For questions 7 through 10, find each sum or difference. Write the result in standard form.

7. $(5g^2h - 3.2gh^2 + 6) + (-2.75gh^2 + 10)$

$5g^2h - 5.95gh^2 + 16$

8. $\frac{1}{2}r^6t^2 - (2r^3t^2 - r^6t^2 + 3rt^2)$

$\frac{3}{2}r^6t^2 - 2r^3t^2 - 3rt^2$

9. $(18a^2 + 5a - 9b^2) + (3a + 6b^2 - 7a^2)$

$11a^2 - 3b^2 + 8a$

10. $(2.7m^2 - 3.1n) - (4n^2 + 5.1m)$

$2.7m^2 - 4n^2 - 5.1m - 3.1n$

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Lesson 6 Practice

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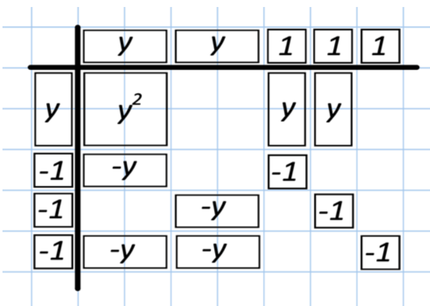
1. Model the following expression with algebra tiles.

$$3x + 5$$

x	x	x	$+1$	$+1$	$+1$	$+1$	$+1$

Use the expression $(2y+3)(y-3)$ for questions 2 through 4.

2. Complete the model to find the product of $(2y+3)(y-3)$ by drawing and labeling the missing tiles.



	y	y	$+1$	$+1$	$+1$
y	y^2	y^2	y	y	y
-1	$-y$	$-y$	-1	-1	-1
-1	$-y$	$-y$	-1	-1	-1
-1	$-y$	$-y$	-1	-1	-1

3. Determine how many of each tile are represented in the product and write the numbers in the table.

y^2 tiles	$+y$ tiles	$-y$ tiles	$+1$ tiles	-1 tiles
2	3	6	0	9

4. Complete the statement.
The product of $(2y+3)(y-3)$ is $2y^2 - 3y - 9$

For questions 5-9, use algebra tiles to find each product.

5. $(2-5x)(x-2)$ = $-5x^2 + 12x - 4$

	$-x$	$-x$	$-x$	$-x$	$-x$	$+1$	$+1$
x	$-x^2$	$-x^2$	$-x^2$	$-x^2$	$-x^2$	x	x
-1	x	x	x	x	x	-1	-1
-1	x	x	x	x	x	-1	-1

6. $(3x+1)^2 = 9x^2+6x+1$

	x	x	x	$+1$
x	x^2	x^2	x^2	x
x	x^2	x^2	x^2	x
x	x^2	x^2	x^2	x
$+1$	x	x	x	1

7. $(x+4)(x-4) = x^2-16$

	x	$+1$	$+1$	$+1$	$+1$
x	x^2	x	x	x	x
-1	$-x$	-1	-1	-1	-1
-1	$-x$	-1	-1	-1	-1
-1	$-x$	-1	-1	-1	-1
-1	$-x$	-1	-1	-1	-1

8. $(x-3)(x+2) = x^2-x-6$

	x	$+1$	$+1$
x	x^2	x	x
-1	$-x$	-1	-1
-1	$-x$	-1	-1
-1	$-x$	-1	-1

9. $(2-x)(3x+1) = -3x^2+5x+2$

	x	x	x	$+1$
$-x$	$-x^2$	$-x^2$	$-x^2$	$-x$
$+1$	x	x	x	$+1$
$+1$	x	x	x	$+1$

10. One of your classmates was absent when you learned how to multiply polynomials using algebra tiles. Describe to your classmate the most important thing* to remember when multiplying with algebra tiles.
Multiplying polynomials using algebra tiles is just like finding the area of a rectangle. Line-up the factors to represent length and width. Fill in the area with blocks that are the same length and width. The blocks in the area space represent the product of the polynomials.

***Be careful with the positive and negative signs.**

Algebra 1
Unit 2
Lesson 7 Practice

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1. Find the product of $(x^2+4)(2x-5)$ using the distributive property.

$2x^3-5x^2+8x-20$

2. Find the product of $(x^2+2)(3x+4)$ using an area model.

	x^2	$+2$
$3x$	$3x^3$	$6x$
$+4$	$4x^2$	8

$3x^3+4x^2+6x+8$

3. Find the product of $(-\frac{1}{2}y+4y^2-3)(4y-5)$ using the distributive property.

$$16y^3-22y^2-\frac{19}{2}y+15$$

4. Find the product of $(\frac{1}{3}y+2y^2-6)(y-4)$ using an area model.

	$\frac{1}{3}y$	$2y^2$	-6
y	$\frac{1}{3}y^2$	$2y^3$	$-6y$
-4	$\frac{-4}{3}y$	$-8y^2$	24

$$2y^3-\frac{23}{3}y^2-\frac{22}{3}y+24$$

Find each product.

5. $(-4a + \frac{3}{5}b)^2$

$$16a^2 - \frac{24}{5}ab + \frac{9}{25}b^2$$

6. $(2.2 - 5c)(0.7 + 4c^3 - c^2)$

$$-20c^4 + 13.8c^3 - 2.2c^2 - 3.5c + 1.54$$

7. $(\frac{3}{5}x - 2)(\frac{-1}{2} + 4x)$

$$\frac{12}{5}x^2 - \frac{83}{10}x + 1$$

8. $(-4\frac{1}{2} + 4d^2)(5d - 7.1c - 2d^2)$

$$-22.5d + 31.95c + 9d^2 + 20d^3 - 28.4cd^2 - 8d^4$$

9. $(14 + 4m)(-6n - \frac{14}{5})$

$$-84n - 24mn - \frac{196}{5} - \frac{56}{5}m$$

10. $(3r^2 - \frac{5}{4} + 15r)(-2r + 3r^2 + 12)$

$$9r^4 + 39r^3 + \frac{9}{4}r^2 + \frac{365}{2}r - 15$$

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1. Rewrite the expression in each step using the description: $6(x+4)+3x$

$6x+24$	Distribute the 6 to $(x+4)$.
$6x+24+3x$ $9x+24$	Combine like terms.

2. Rewrite the expression in each step using the description: $-3x+5(-2x-5)$

$-10x-25$	Distribute the 5 to $(-2x-5)$.
$-3x-10x-25$ $-13x-25$	Combine like terms.

Use the order of operations to simplify each polynomial.

3. $3(x+5)+2(x-8)$

$3x+15+2x-16$

$5x-1$

4. $\frac{3}{5}(x-6)+3x$

$$\frac{3}{5}x-\frac{18}{5}+3x$$

$$\frac{18}{5}x-\frac{18}{5}$$

5. $-2(x^2+4x)+7(x-2)$

$$-2x^2-8x+7x-14$$

$$-2x^2-x-14$$

6. $2x^2-45(x-7)$

$$2x^2-45x+315$$

7. $(x+3)(x-5)-9(2x^2+3x)$

$$x^2-5x+3x-15-18x^2-27x$$

$$-17x^2-29x-15$$

8. $-\frac{1}{3}(6x-12)+7x$

$$-2x+4+7x$$

$$5x+4$$

9. $2x^2+(x+5)(x-5)$

$$2x^2+x^2-5x+5x-25$$

$$3x^2-25$$

10. $3x(x-4)+7(x-3)$

$$3x^2-12x+7x-21$$

$$3x^2+-5x-21$$

Algebra 1
Unit 2
Lesson 9 Practice

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1. Use the area model to find $(2.4y^3 + 1.4x^3y^2 - 3.2x^2y) \div (2xy)$ where $x \neq 0$ and $y \neq 0$.

$\frac{1.2y^2}{x} + 0.7x^2y - 1.6x$

	$\frac{1.2y^2}{x}$	$0.7x^2y$	$-1.6x$
$2xy$	$2.4y^3$	$1.4x^3y^2$	$-3.2x^2y$

$\frac{6y^2}{5x} + \frac{7x^2y}{10} - \frac{8x}{5}$

Find each quotient.

2. $(\frac{1}{4}x^3 + \frac{1}{3}x^2) \div (-\frac{1}{4}x^2)$ where $x \neq 0$

$-x - \frac{4}{3}$

3. $(5a^3 + 10a^2 - 30a) \div (5a)$ where $a \neq 0$

$a^2 + 2a - 6$

4. $(6k^6r^5 + 3k^3r^4 - 12k^3r^2) \div (3k^2r)$ where $k \neq 0$ and $r \neq 0$

$2k^4r^4 + kr^3 - 4kr$

5. $(-24w^6 + 12w^3 - 16w^2) \div (4w)$ where $w \neq 0$

$-6w^5 + 3w^2 - 4w$

6. $\frac{14x^4-42x^3+21x^2-28}{7x}$ where $x \neq 0$

$2x^3 - 6x^2 + 3x - \frac{4}{x}$

7. $\frac{12p^4+42p^3+30p^2-120p}{6p}$ where $p \neq 0$

$2p^3 + 7p^2 + 5p - 20$

8. $\frac{4t^4-8t^3+12t^2-8t}{4t}$ where $t \neq 0$

$t^3 - 2t^2 + 3t - 2$

9. $\frac{3.5x^3+4.5+3.5x^2-0.5}{0.5x}$ where $x \neq 0$

$7x^2 + 7x + \frac{8}{x}$

10. $\frac{16y^3+46y^2+40y^2}{-4y^3}$ where $y \neq 0$

$-4 - \frac{43}{2y}$